

Toy Search Engine

- a search engine is an info. retrieval sys. Typically it is used to locate info. from text or some databases.
- In this we will see a toy search engine where we mean to search on a collection of 5 documents.
- Any set of documents

↓
Tokenize your text (white space based)

- words are atomic units of meaning
- Tokens → are individual occurrences
of words or symbols in a text.
- Types → are unique words or
symbols in a text.

↓ some more pre-processing steps

- spell check
- suffix stripping
- stopwords removal

↓
inflection → modifies a word to
express grammatical info.
w/o changing its core meaning
or word class.

↓
→ derivation → creates a new word by adding prefix or suffix, often changing the meaning & word class.

after spellcheck

Stemming is chopping off the word endings using simple rules w/o understanding the word meaning.

Lemmatization it reduces the word to its dictionary form (lemma) using vocab & grammar rules

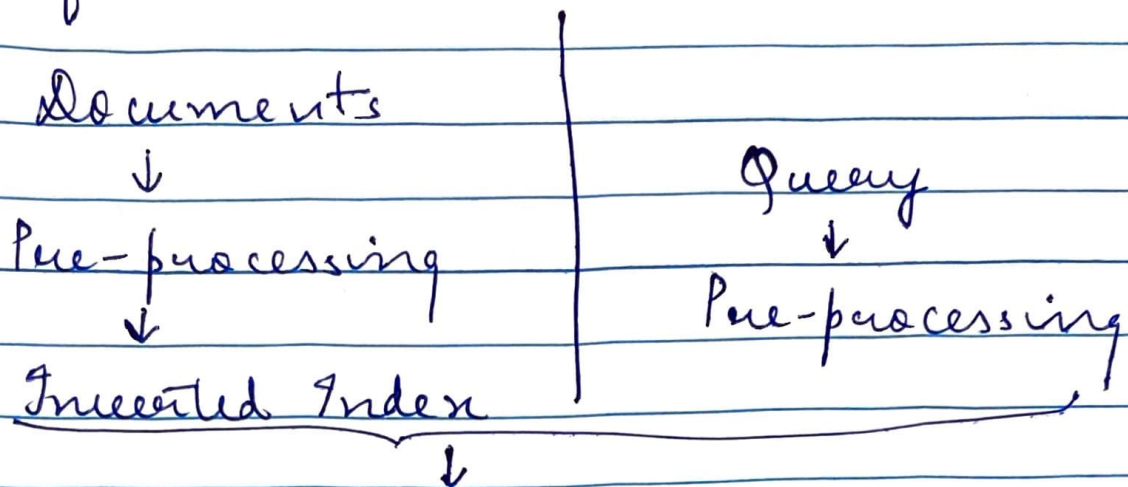
↓
stop word removal

→ generally the most common words in a language. No single universal list. It depends on the corpus & given task at hand.

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Inverted Indexing

↪ kind of data structure that maps the words to the documents that contain them.

- so far,



Then, retrieval from the inverted index.

Then collect the retrieved documents

give ranking to them based on the max. count.

- IDF $\rightarrow$ measures how rare & imp. a word is across the collection of documents.

$$IDF(t) = \log \left(\frac{N}{df(t)} \right)$$

$t \rightarrow$ term

$N \rightarrow$ total no. of docs.

$df(t) \rightarrow$ no. of docs having term t .

provides
a
discriminating
power.